B1.2 Proteins (HL)

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

9 MARKS

1 STATE two types of R-group found in amino acids. [B1.2.6] [2]

2 NAME the amino acid that forms disulfide bonds. [B1.2.9] [1]

3 STATE which groups in the backbone form hydrogen bonds in secondary structure. [B1.2.8] [1]

4 DEFINE conjugated protein. [B1.2.11] [1]

5 STATE the number of polypeptide chains in insulin, collagen and haemoglobin. [B1.2.11] [3]

6 STATE one difference in shape between globular and fibrous proteins. [B1.2.12] [1]

Section B: Understanding

15 MARKS

7 EXPLAIN why the primary structure of a protein determines its three-dimensional shape. [B1.2.7] [3]

8 EXPLAIN how a change in pH can disrupt the tertiary structure of a protein. [B1.2.9] [3]

9 **EXPLAIN** where hydrophobic and hydrophilic amino acids are found in an integral membrane protein. [3]
[B1.2.10]

10 **EXPLAIN** how the structure of collagen suits its function. [B1.2.12] [3]

11 **EXPLAIN** how the chemical variety of R-groups produces a great diversity of proteins. [B1.2.6] [3]

Section C: Working with evidence

10 MARKS

All data in this section are invented for practice. The patterns are realistic.

12 [B1.2.7, B1.2.9]

An enzyme with one chain of 124 amino acids and eight cysteines was unfolded with urea (which breaks hydrogen bonds and hydrophobic interactions) and a reducing agent (which breaks disulfide bonds). Its activity was then measured under different conditions.

SAMPLE	TREATMENT	ACTIVITY / % OF UNTREATED
A	Untreated	100
B	In urea and reducing agent	2
C	B, then urea and reducing agent both removed slowly	92
D	B, then reducing agent removed first and urea an hour later	3
E	D, then a trace of reducing agent added for ten hours	88

a **EXPLAIN** why sample B had almost no activity. [2]

b **DEDUCE** what sample C shows about protein structure. [2]

c **SUGGEST** why sample D stayed inactive. [2]

d **SUGGEST** how adding a trace of reducing agent restored activity in E. [2]

e **STATE** the largest number of disulfide bonds this enzyme could have. [1]

f **STATE** which level of protein structure is not affected by urea. [1]

Section D: Longer answer

6 MARKS

13 **COMPARE AND CONTRAST** the structure and function of insulin and collagen. [B1.2.11, B1.2.12] [6]